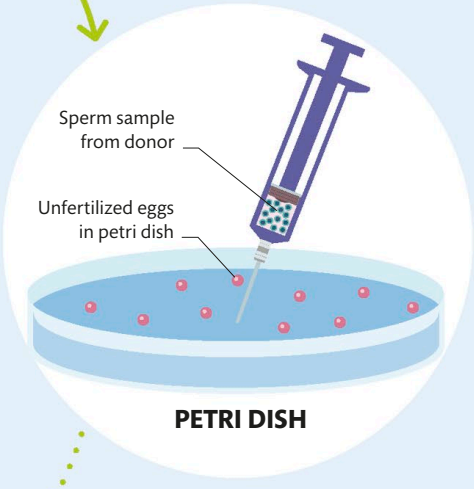
Assisted fertility

Assisted fertility techniques are used to help people conceive a healthy baby. The most common methods are intrauterine insemination (IUI) and in vitro fertilization (IVF), or "test-tube" fertilization.

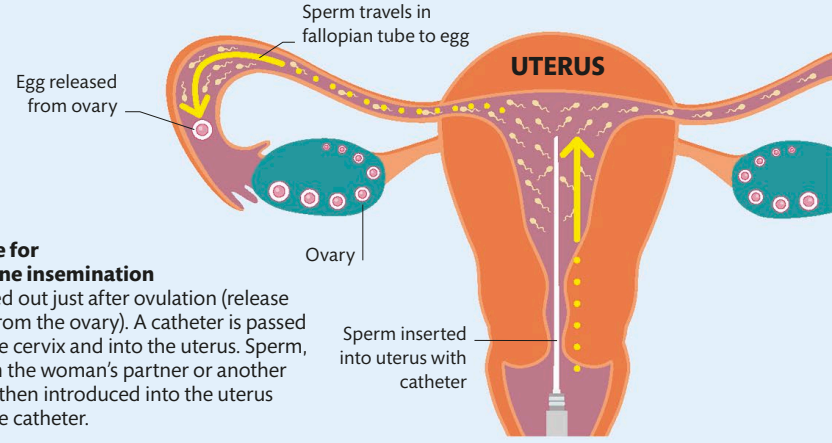
Procedure for IVF
Eggs are collected from the woman and sperm from the man. The sperm and eggs are brought together in a laboratory. Alternatively, a sperm is injected into an egg to ensure fertilization, a techique known as intracytoplasmic sperm injection, or ICSI (see panel, below right). The fertilized egg (called an embryo) is then introduced into the uterus, where it may implant in the lining of the uterus.



3 Sperm combined with eggs
The eggs are checked for quality then mixed with sperm and incubated at body temperature (98.6°F/37°C) in a petri dish. The mixture is checked the next day to see if any eggs have been fertilized.

Intrauterine insemination

Normally, fertilization occurs when a sperm fuses with an egg in the fallopian tube after sexual intercourse. The resulting fertilized cell then implants in the lining of the uterus to become an embryo. In intrauterine inseminination, sperm are introduced into the uterus through a catheter (a thin, hollow tube). IUI may be recommended if the woman cannot become pregnant naturally, if the man has insufficient healthy sperm, or if donated sperm are used.



Procedure for intrauterine insemination
IUI is carried out just after ovulation (release of an egg from the ovary). A catheter is passed through the cervix and into the uterus. Sperm, either from the woman's partner or another donor, are then introduced into the uterus through the catheter.

HOW DOES AGE AFFECT FERTILITY?

After about the mid-20s, a woman's fertility declines with age, with the biggest decrease occurring from the mid-30s. Male fertility also declines from about the 20s but less sharply.

ICSI

In intracytoplasmic sperm injection (ICSI), a man gives a sperm sample, and a single, healthy sperm cell is selected. This sperm is then injected directly into an egg removed from the woman. ICSI is usually done if a man has too few sperm or very few healthy sperm.

